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High-power continuous-wave optical waveguiding in a semiconductor nanowire

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Abstract

Semiconductor nanowires have been extensively studied for a wide range of nanophotonic applications, including nanolasers, photodetectors, nonlinear frequency converters, and solar cells. While high-power optical waveguiding is often desired in these applications, the guided power (or average power) in individual nanowires has so far been typically limited to about 100 μW due to propagation losses and low-efficiency optical coupling. Here, we demonstrate high-power continuous-wave (CW) optical waveguiding in CdS and ZnO nanowires enabled by efficient in- and out-coupling using silica fiber tapers. A single 550-nm-diameter CdS nanowire at 1550-nm wavelength is shown to safely guide optical power up to 2.5 W, which is approximately four orders of magnitude higher than previous results. Using these high-power CW optical waveguiding nanowires, we further realize efficient nonlinear frequency conversion with second- and third-harmonic generation efficiencies of 6.9×10^{-6} and 6.4×10^{-8}, respectively, comparable to those obtained under pulsed excitation. These results may push nanowire photonics into the high-power regime and open up new opportunities for both fundamental studies and technological applications.

1 | Introduction

Semiconductor nanowires have emerged as ideal platforms for investigating light-matter interactions at the nanoscale owing to their high aspect ratio, strong optical confinement, large evanescent field, and tunable bandgap[1-3]. These properties enable them to serve as versatile nanoscale building blocks for diverse photonic and optoelectronic applications, including nanolasers[4-6], micro-LEDs[7-9], photodetectors[10-12], nonlinear frequency converters[13, 14] and solar cells[15, 16]. A direct and effective approach to enhancing light-matter interactions and exploring new technological opportunities is to increase the optical power that can be guided within the nanowires. However, the maximum guided power achieved in individual nanowires has so far been typically limited to about 100 μW[17-19], primarily due to loss-related limitations such as absorption-induced heating, scattering-induced signal degradation, as well as inefficient optical coupling.

Typical semiconductor nanowires, such as cadmium sulfide (CdS) and zinc oxide (ZnO) nanowires, feature atomically smooth surfaces[1], low waveguiding loss[20], high damage thresholds[21-23], and large surface-to-volume ratios that facilitate efficient heat dissipation[24]. When operated in weak-absorption spectral regions (e.g., CdS exhibits an ultralow absorption coefficient at a wavelength of 1550 nm)[25], these nanowires are theoretically capable of supporting guided optical powers far exceeding those reported to date. For example, the theoretical tolerable average power density in CdS nanowires is expected to approach that of bulk CdS crystals ($\sim 6.5\times10^{13}$ W/m^2)[26]. However, such high power densities have not yet been experimentally realized in nanowire waveguides.

Here we demonstrate high-power continuous-wave (CW) optical waveguiding in CdS and ZnO nanowires enabled by an in-line fiber-nanowire-fiber coupling configuration using silica fiber tapers[27, 28]. A single 550-nm-diameter CdS nanowire at a wavelength of 1550 nm is shown to safely guide CW optical power up to 2.5 W, corresponding to a power density of 1.1×10^{13} W/m^2. This represents an improvement of approximately four orders of magnitude over previously reported nanowire waveguides and approaches, to our knowledge, the highest experimentally reported power density for nondestructive operation in bulk CdS crystals[26]. In addition, a 690-nm-diameter ZnO nanowire operating at 532 nm achieves a guided power exceeding 33 mW (8.8×10^{10} W/m^2), which is about two orders of magnitude higher than previously reported values[17-19]. Using these high-power waveguiding nanowires, we furthermore realize efficient

CW nonlinear frequency conversion with efficiencies of 6.9×10^{-6} and 6.4×10^{-8} for second-harmonic generation (SHG) and third-harmonic generation (THG), respectively.

2 | Results

2.1 | Optical characterization of nanowires

The CdS and ZnO nanowires used in this work were synthesized via chemical vapor deposition (CVD)[29, 30]. The morphology and crystal structure of the as-grown nanowires were characterized by scanning electron microscopy (SEM) and high-resolution transmission electron microscopy (HR-TEM). As shown in Figure 1a, a representative 550-nm-diameter CdS nanowire exhibits a uniform diameter and an atomically smooth surface. HR-TEM imaging in Figure 1b confirms its single-crystalline structure with a surface roughness below 0.1 nm (see Figure S1). These high-quality structural characteristics are critical for achieving low propagation loss during optical waveguiding. Similarly, SEM and HR-TEM images of a ZnO nanowire also confirm its high quality with atomically smooth surfaces and single-crystalline structures (Figures 1c-d).

The defect states of individual nanowires were examined by photoluminescence (PL) spectroscopy after transferring the nanowires onto a MgF_2 substrate. Under excitation with 355-nm laser pulses (1 ns pulse duration, 1 kHz repetition rate), the ZnO nanowire exhibits both near-band-edge emission and a broad defect-related band centered at ~550 nm (Figure 1e), consistent with oxygen vacancy–related states reported previously[31, 32]. Despite the presence of defect-related emission, the ZnO nanowires still maintain sufficient optical quality to support high-power waveguiding in the visible regime. In contrast, the CdS nanowire shows only clean band-edge emission without detectable defect-related luminescence, indicating a low density of crystal defects. The absence of the defect states in CdS is particularly important for suppressing absorption-induced heating and enabling high-power continuous-wave waveguiding.

The optical waveguiding properties of CdS nanowires were further investigated using dark-field fluorescence microscopy under 405-nm CW laser excitation. As shown in Figure 1f, for excitation at different positions along the nanowire, the emitted light is predominantly confined to the distal ends, with negligible scattering observed along the nanowire body. This behavior confirms the excellent optical quality and low propagation loss of the nanowires, which is essential for achieving high-power optical waveguiding in the nanowires.

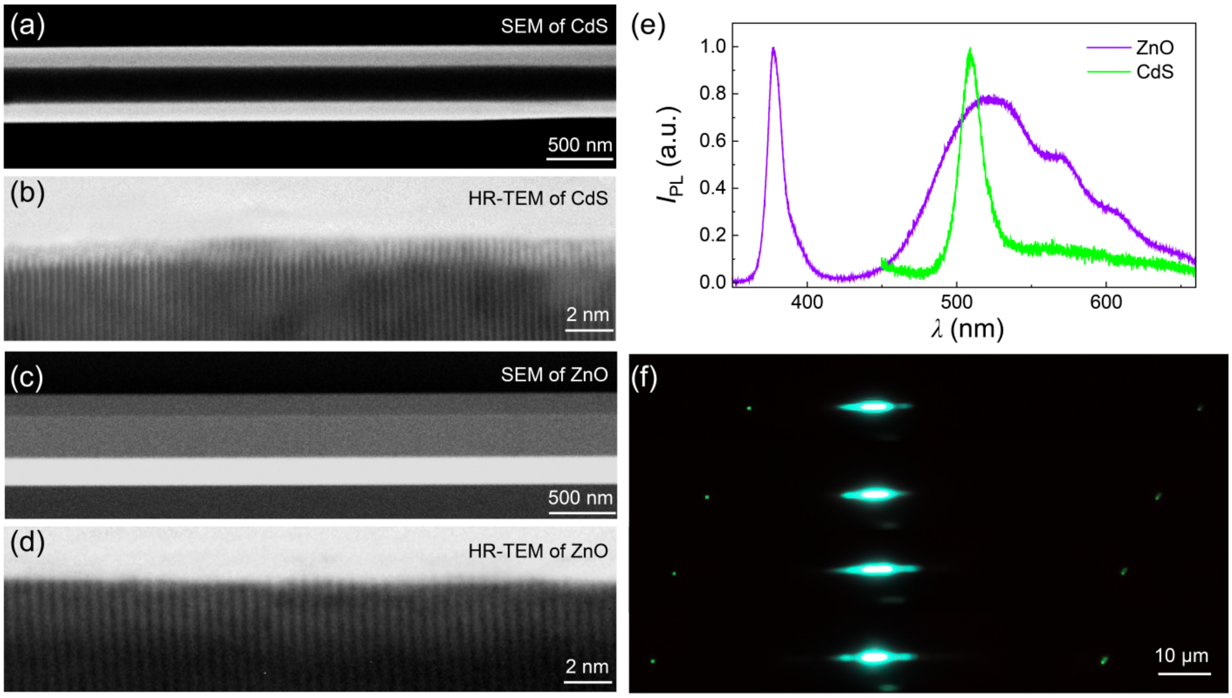

Figure 1. Optical characterization of nanowires. (a) Scanning electron microscopy (SEM) image of a representative 550-nm-diameter CdS nanowire. (b) High-resolution transmission electron microscopy (HR-TEM) image of a 130-nm-diameter CdS nanowire. (c) SEM image of a representative 690-nm-diameter ZnO nanowire. (d) HR-TEM image of a 480-nm-diameter ZnO nanowire. (e) Photoluminescence spectra (I_{PL}) of a 940-nm-diameter ZnO (purple line) and a 600-nm-diameter CdS (green line) nanowire, respectively. The ZnO spectrum exhibits a broad defect-related peak centered at ~550 nm in addition to its band-edge emission, attributed to oxygen vacancies. (f) Dark-field photoluminescence microscopy images of a CdS nanowire with the excitation spot of a 405 nm CW laser scanned along its length. PL emission is predominantly observed at the distal ends, with negligible scattering along the nanowire.

2.2 | High-power CW optical waveguiding in nanowires

The transmission of high-power CW light in nanowires is mainly limited by absorption-induced thermal damage and low coupling efficiency. To mitigate these effects and investigate the high-power optical waveguiding behavior, we chose input light within the weak-absorption regime of the nanowires and employed an in-line fiber-nanowire-fiber coupling configuration for efficient light input and output. As schematically illustrated in Figure 2a, the input light from a standard optical fiber was adiabatically squeezed into a fiber taper drawn at its end, and then evanescently coupled into a suspended nanowire. The transmitted light at the opposite end was collected by a second tapered fiber and measured with a power meter. The fiber tapers were fabricated from standard single-mode fiber using a flame-heated taper drawing technique to satisfy the adiabatic criterion.

During the experiment, an individual nanowire was brought into side-by-side contact with the fiber taper under an optical microscope, and the coupling was optimized via micromanipulation to maximize the output. Figure 2b shows typical optical microscope images of the evanescent coupling between a fiber taper and individual nanowires. For a 550-nm-diameter CdS nanowire, the bright-field image (Figure 2b, top) presents the coupling structure, while the corresponding dark-field image (Figure 2b, middle) clearly displays guided 1550-nm light propagating through the structure. The negligible scattering along the CdS nanowire, combined with a strong emission solely from its end facet, confirms highly efficient CW-light coupling from the fiber taper, with the coupling efficiency exceeding 90% (see Figure S2). Similarly, efficient coupling is also obtained for a 690-nm-diameter ZnO nanowire, as demonstrated by guided 532-nm CW light in the dark-field image (Figure 2b, bottom).

To investigate the high-power waveguiding properties, 1550 nm CW light from a fiber laser was amplified using an erbium-doped fiber amplifier (EDFA, Connet, MFAS-Er-C-B-15) and coupled into the nanowire. The output was collected by a power meter, and the guided power along the nanowire was also obtained by evaluating the coupling efficiency between the nanowire and the fiber tapers (see Figure S3). Considering possible hysteresis response of the system induced by thermal effects, we swept the input power back and forth from minimum to maximum. As shown in Figure 2c, the measured output power P_{out} changes linearly with the input power P_{in} in a 550-nm-diameter CdS nanowire without observable hysteretic effect. An output power of up to 1.8 W was achieved, corresponding to a guided power of 2.5 W and a power density of 1.1×10^{13} W/m² along the nanowire (see Supplementary text 2 for waveguiding power calculations). This guided power is nearly four orders of magnitude higher than previously reported values (~100 μW[17-19]), with the achieved power density approaching, to our knowledge, the highest experimentally reported nondestructive power density in bulk CdS crystals (~6.5×10^{13} W/m²)[26]. The possibility of waveguiding such high optical power along a nanowire can be attributed to the highly efficient coupling, low propagation loss, and high damage threshold of the nanowire. Moreover, the power-dependent temperature rise of the nanowire during high optical power waveguiding was further evaluated using the Varshni effect[33, 34]. The estimated nanowire temperature reaches approximately 180 °C at a guided power of 1.1 W, which remains far below the known thermal damage threshold of the CdS crystals (see Figure S4).

The high-power waveguiding characteristics of ZnO nanowires were also examined (Figure 2d). For a 690-nm-diameter ZnO nanowire, the maximum measured output power reached 231 mW at 1550 nm and 17 mW at 532 nm, corresponding to a guided power of 292 mW at 1550 nm and 33 mW at 532 nm. Compared to CdS nanowires, the lower guided power is mainly constrained by the intrinsic damage threshold of bulk ZnO (~10^{15} W/m^2)[23], which is slightly lower than that of bulk CdS (~10^{16} W/m^2)[21, 22]. At 532 nm, the output power is further limited by strong optical absorption associated with oxygen-vacancy-related defect states, as indicated in Figure 1e.

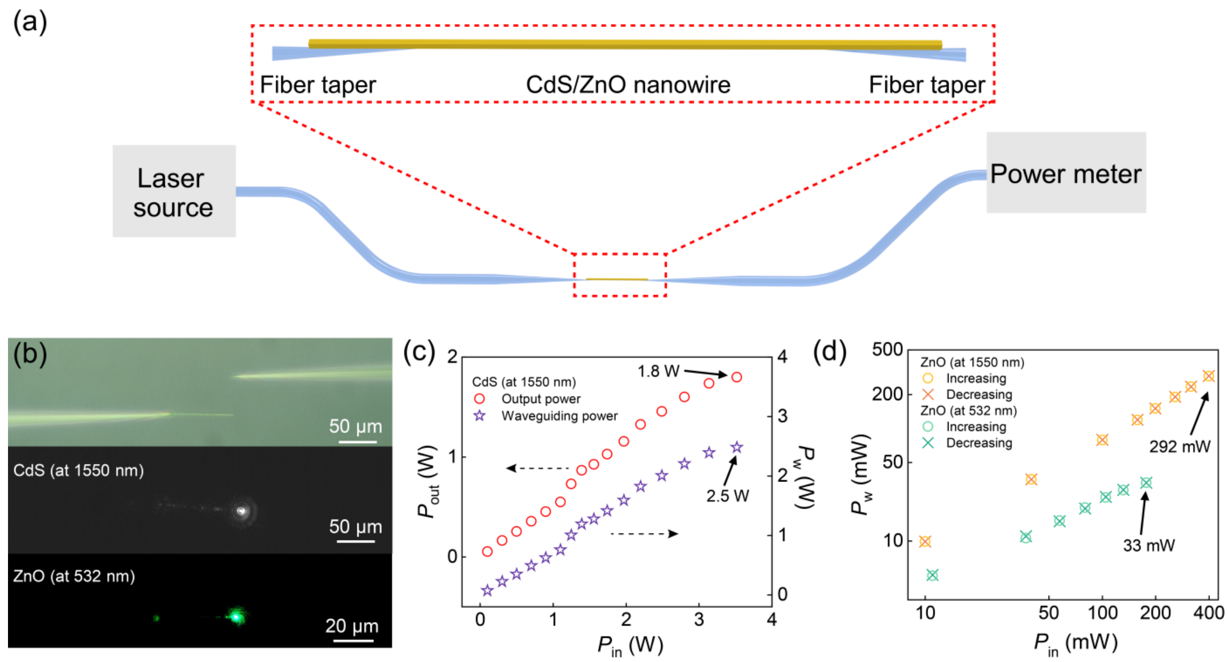

Figure 2. High-power CW optical waveguiding in single nanowires. (a) Schematic diagram of experimental setup for investigating high-power CW optical waveguiding in a nanowire. EDFA, erbium-doped fiber amplifier. (b) Optical microscope images of evanescent coupling between fiber tapers and nanowires. (Top) Bright-field and (Middle) dark-field images of a 550-nm-diameter CdS nanowire waveguiding 1550 nm light. (Bottom) Dark-field image of a 690-nm-diameter ZnO nanowire waveguiding 532 nm light. (c) Output power (P_{out}, red circle) and calculated waveguiding power (P_w, purple pentagram) as a function of input power (P_{in}) for a 550-nm-diameter CdS nanowire at 1550 nm, respectively. (d) Waveguiding power versus input power for a 690-nm-diameter ZnO nanowire at 1550 nm (yellow) and 532 nm (green). For each

wavelength, waveguiding power was measured during both increasing (circles) and decreasing (crosses) input power sweeps.

2.3 | Diameter-dependent high-power waveguiding in CdS nanowires

For a given wavelength and nanowire morphology, the supported waveguide modes and effective refractive index are primarily determined by the nanowire diameter[35]. Consequently, the diameter critically determines both transmission loss and the coupling efficiency of the fiber-nanowire-fiber structure, and thus directly limits the achievable output power. To explore this effect, we investigated high-power waveguiding properties of CdS nanowires with diameters ranging from 100 nm to 1.1 μm (Figure 3). For all measured diameters, the output power P_{out} increases linearly with the input power P_{in}. However, watt-level output power is achieved only for nanowires whose diameters meet both the single-mode transmission requirement and effective refractive index matching condition (e.g., 280 nm and 550 nm). Nanowires with diameters outside the optimal single-mode range (approximately 260 nm to 600 nm) experience additional losses due to undesirable mode leakage or multimode propagation (see Figure S5), and also exhibit reduced coupling efficiency arising from effective refractive index mismatch between the nanowires and the fiber tapers. Therefore, the appropriate diameter selection is essential for realizing high-power, single-mode nanowire waveguides.

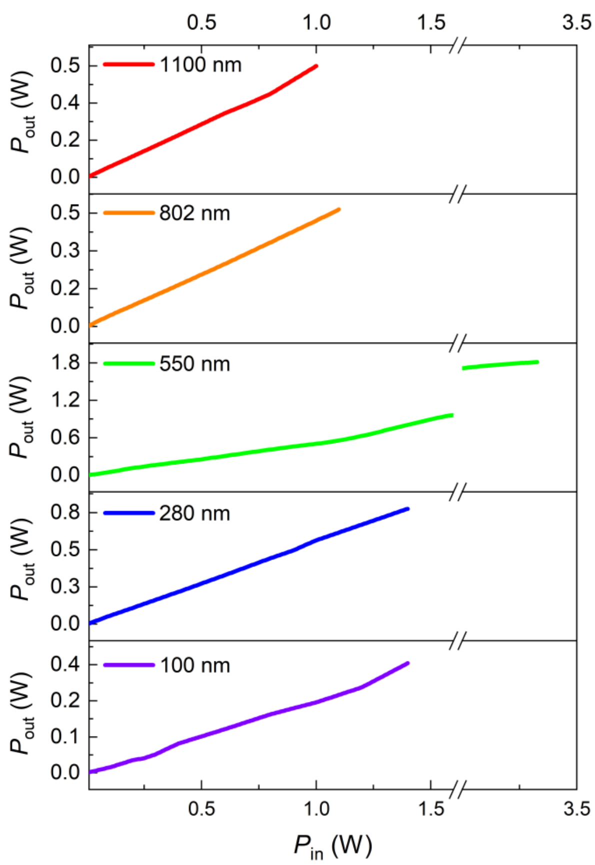

Figure 3. Diameter-dependent high-power waveguiding in CdS nanowires at 1550 nm. Measured output power (P_{out}) as a function of the input power (P_{in}) for CdS nanowires with diameters of 100 nm (purple), 280 nm (blue), 550 nm (green), 808 nm (orange), and 1100 nm (red).

2.4 | Nonlinear harmonic generation in a high-power CW waveguiding CdS nanowire

Semiconductor nanowires provide an attractive platform for nanoscale nonlinear optics owing to their strong optical confinement, large optical nonlinearity, and tailorable waveguide dispersion. High pump power in the nanowire is usually desired to enhance effective nonlinear light-matter interactions and the associated nonlinear optical effects. Based on the demonstration of watt-level CW waveguiding in CdS nanowires, we further investigated CW nonlinear effects, particularly the SHG and THG in the nanowires.

Experimentally, a 550-nm-diameter, 70-μm-long CdS nanowire was pumped with 1550-nm CW light. Clear SHG at 776 nm and THG at 520 nm were observed, as shown in Figure 4a, with strong SHG emission emerging from the output facet of the nanowire. To optimize the intermodal phase matching, a tunable laser (Santec, TSL-710) was employed as the seed source for fine wavelength tuning, and the generated harmonic signals were collected and analyzed using a spectrometer (HORIBA Jobin Yvon, iHR550). As shown in Figures 4b and 4c, for a representative 550-nm-diameter, 70-μm-long CdS nanowire waveguiding a 0.2-W-power CW light, the SHG and THG intensities are maximized at pump wavelengths of 1551 nm and 1562 nm, respectively, indicating efficient phase-matched enhancement.

To quantify the frequency conversion efficiencies in the CW waveguiding CdS nanowire, we measured the dependence of the SHG and THG output on the guided power by increasing the input power under phase matching conditions (Figure 4d). Both harmonic signals exhibit linear behavior in a log-log plot, yielding fitted slopes of 1.97 ± 0.14 for SHG and 3.2 ± 0.18 for THG, which agree well with the theoretical quadratic and cubic dependencies for SHG and THG, respectively. At a guided power of 1.1 W, the frequency conversion efficiencies reach 6.9×10^{-6} for SHG and 6.4×10^{-8} for THG, comparable to previously reported values obtained under pulsed excitation[36-38]. These high CW conversion efficiencies can be mainly attributed to the high optical power density within the nanowires and the well-achieved phase matching.

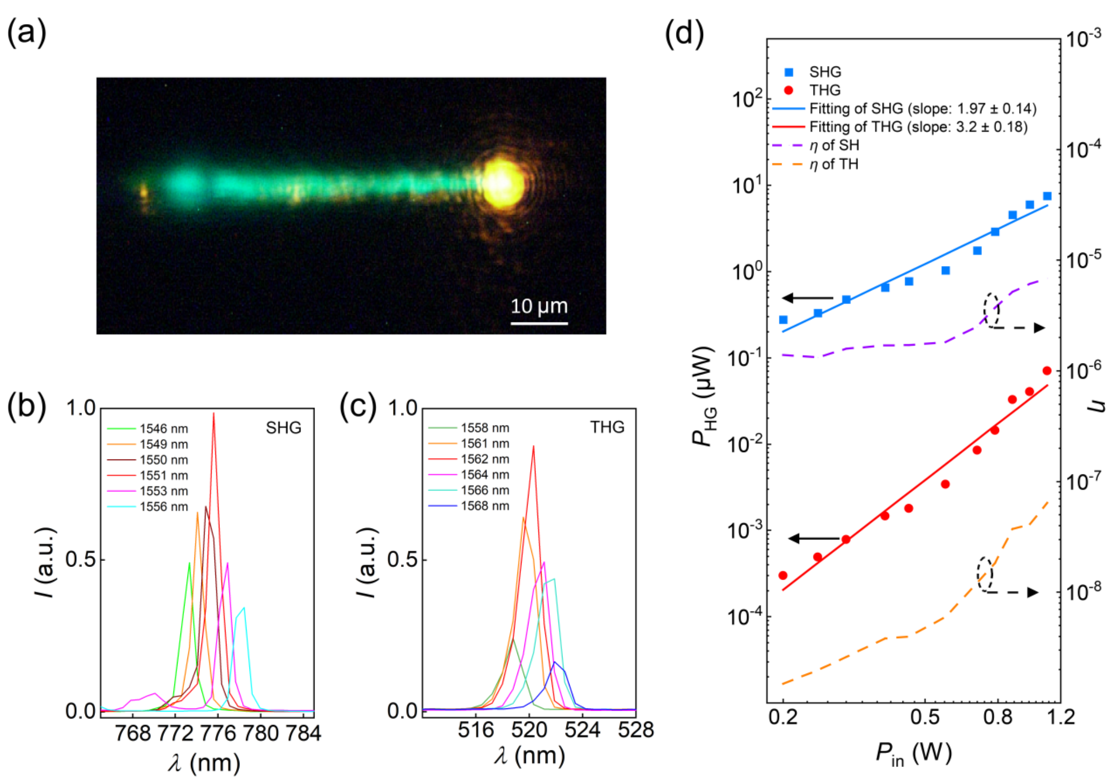

Figure 4. Harmonic generation in a high-power CW waveguiding CdS nanowire. (a) Dark-field optical microscope image of the THG and SHG from a 550-nm-diameter, 70-μm-long CdS nanowire waveguiding a 0.2-W-power CW light. (b and c) Corresponding measured (b) SHG and (c) THG spectra. The pump wavelength was tuned across a range of 1546-1556 nm for SHG and 1558-1568 nm for THG to optimize intermodal phase matching. (d) Dependence of the harmonic generation power (P_{HG}) and conversion efficiencies (η) on the input power (P_{in}) for SHG and THG. Linear fits on log-log plots yield slopes of 1.97 ± 0.14 and 3.2 ± 0.18 for SHG and THG, respectively. The maximum conversion efficiencies are 6.9×10^{-6} for SHG and 6.4×10^{-8} for THG with a 1.1-W CW light guided in the CdS nanowire.

3 | Conclusion

In summary, we have demonstrated high-power CW optical waveguiding in CdS and ZnO nanowires. A single 550-nm-diameter CdS nanowire safely waveguides 1550-nm CW light with a power up to 2.5 W, nearly four orders of magnitude higher than previously reported results. Similarly, a 690-nm-diameter ZnO nanowire supports guided powers of 292 mW at 1550 nm and 33 mW at 532 nm. Enabled by this high-power CW waveguiding, we further realized intermodal-phase-matched second- and third-harmonic generation, achieving nonlinear frequency conversion efficiencies comparable to those obtained under pulsed excitation. As high-power CW waveguiding is desired for nanowire-based photonic applications, these results may push the semiconductor nanowire photonics into the high-power regime and open new opportunities for applications ranging from nanolasers and photodetectors to nonlinear optics.

Author contributions

Hongliang Dang: writing – original draft, validation, methodology, software, formal analysis, data curation, and conceptualization. **Zhanke Zhou:** validation, methodology, investigation and formal analysis. **Jianbin Zhang:** investigation and formal analysis. **Liu Yang:** investigation and formal analysis. **Xin Guo:** writing – review & editing, supervision, resources, funding acquisition, project administration, and conceptualization. **Limin Tong:** writing – review & editing, supervision, resources, funding acquisition, project administration, and conceptualization.

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Conflicts of Interest

The authors declare no competing financial interest.

Data Availability Statement

The data that support the findings of this study is available within the article and its Supporting Information.

List of Non-Standard Abbreviations

CW, Continuous-wave; SHG, second-harmonic generation; THG, third-harmonic generation; CdS, Cadmium Sulfide; ZnO, Zinc Oxide; CVD, chemical vapor deposition; SEM, scanning electron microscopy; HR-TEM, high-resolution transmission electron microscopy; PL, photoluminescence; EDFA, erbium-doped fiber amplifier.

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Supporting Information

Detailed information on the measurement of surface roughness, simulation of coupling processes and waveguiding power, round-trip measurements of high-power waveguiding properties, evaluation of nanowire temperature utilizing the Varshni effect, and mode analysis of the CdS nanowire (PDF).

High-power continuous-wave optical waveguiding in a semiconductor nanowire

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S1 | Measurement of surface roughness of a CdS nanowire

We used an image recognition algorithm to analyze the high-resolution TEM images of the CdS nanowires and extract the lattice boundary coordinates of the nanowire sidewalls (Figure S1a). The root-mean-square (RMS) surface roughness of the nanowires was then obtained using the least-squares method to be ~0.1 nm, indicating that the nanowires possess atomic-scale surface smoothness.

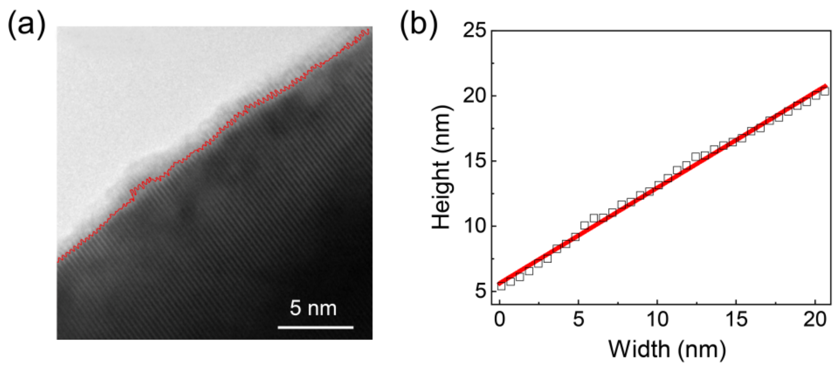

Figure S1. Measured surface roughness of a CdS nanowire. (a) High-resolution TEM image of the sidewall of a 550-nm-diameter CdS nanowire, with the boundary marked by red line. (b) Surface boundaries of the CdS nanowire measured (black rectangle) by comparing the grayscale of the HR-TEM image in (a) and fitted (red line) using the least squares method, from which we obtain a root-mean-square surface roughness of ~0.1 nm.

S2 | Simulation of coupling processes and waveguiding power in nanowires

Using finite-difference time-domain (FDTD) simulations, we investigated the typical near-field coupling process between a 5° fiber taper and a 550-nm-diameter CdS nanowire at a wavelength of 1550 nm. Figures S2a and S2b show the calculated optical power density distributions for coupling light from the fiber taper to the nanowire and from the nanowire to the fiber taper, respectively. Efficient coupling was achieved in both directions, with simulated coupling efficiencies (η) of 92.2% and 93.2%, respectively (Figure S2c).

Given that both values are greater than 90% and nearly identical, we assume equivalent coupling efficiencies for light coupling into and out of the nanowire. Under this assumption, the waveguiding power (P_{w}) along the nanowire can be estimated as

$$P_{\mathrm{w}} = \frac{P_{\mathrm{out}}}{\eta} = P_{\mathrm{out}} \cdot \sqrt{\frac{P_{\mathrm{in}}}{P_{\mathrm{out}}}}, \quad (1)$$

where P_{out} is the output power measured by the power meter (see Figure 2a), P_{in} is the input laser power and $\eta = \sqrt{P_{out}/P_{in}}$ is the coupling efficiency. In practice, experimental uncertainties such as variations in the taper geometry and fluctuations in the effective coupling length may lead to a slightly reduced coupling efficiency compared with the simulated values. Nevertheless, the experimentally measured coupling efficiency (>70%) remains sufficiently high to justify the assumption of bidirectional equivalence for estimating the guided power.

This assumption is not applicable in cases where evident propagation loss or inefficient coupling is present, such as for ZnO nanowire at 532 nm, where increased absorption due to oxygen-vacancy-related defect states leads to higher propagation loss compared with that at 1550 nm. Under such conditions, the actual guided power along the nanowire is expected to be higher than the value estimated using Equation 1.

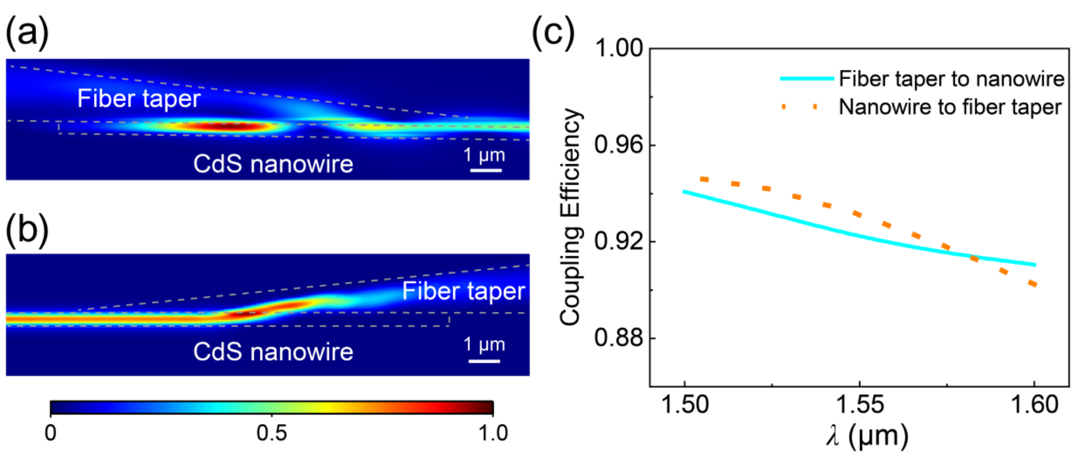

Figure S2. Simulation of near-field coupling processes in a fiber-nanowire-fiber coupling structure. (a, b) Calculated power density distribution of the fundamental TE mode at 1550 nm for coupling light (a) from the fiber taper to the nanowire and (b) from the nanowire to the fiber taper, respectively. (c) Calculated bidirectional coupling efficiencies between the fiber taper and the CdS nanowire with TE polarized input modes.

S3 | Round-trip measurement of high-power waveguiding properties

To investigate the possible thermally induced hysteretic response, we measured the waveguiding characteristics of the nanowires by sweeping the input power forward and backward. As shown in Figure S3a, the measured waveguiding power (P_w) in a 540-nm-diameter nanowire at 1550 nm wavelength changes linearly with the input power (P_{in}) with no observable hysteresis, which also shows the stability of the fiber-nanowire-fiber coupling structure. Figure S3b presents the corresponding measurements for a 690-nm-diameter ZnO nanowire at 1550 nm, confirming a similarly hysteresis-free behavior.

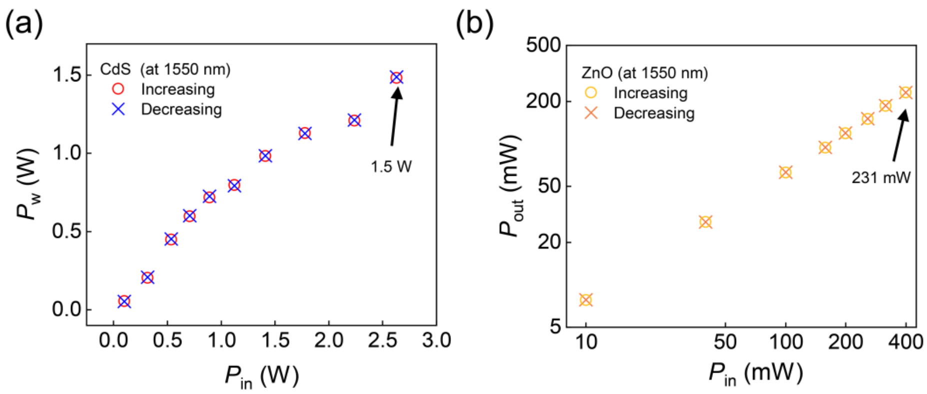

Figure S3. Waveguiding characteristics measured via round-trip input power sweeps. (a) Waveguiding power (P_w) along the nanowire as a function of the input power (P_{in}) for a 540-nm-diameter CdS nanowire at 1550 nm. (b) Output power (P_{out}) versus input power for a 690-nm-diameter ZnO nanowire at 1550 nm. Circles and crosses correspond to increasing and decreasing input power, respectively.

S4 | Evaluation of the temperature of a nanowire by the Varshni effect

To investigate the temperature rising in a high-power waveguiding nanowire, we employed the Varshni effect, an empirical relationship between a material's bandgap and its temperature, using the following formula[1, 2]

$$E_g(T) = a_0 - \frac{\alpha T^2}{\beta + T}, (2)$$

where a_0, α, and β are the Varshni thermal coefficients related to given materials. Experimentally, we placed a single CdS nanowire on a MgF_2 substrate, which was mounted on the top of an electrically heated hotplate with temperature precisely controlled. We excited the nanowire using a 355 nm nanosecond laser (Crystal laser, 1Q 355-4; 1 kHz, 1 ns), and collected its photoluminescence spectrum with a spectrometer (HORIBA Jobin Yvon, iHR550). As shown in Figure S4a, the measured photoluminescence peak of the nanowire exhibits a monotonic red shift with increasing temperature. This shift was fitted using the Varshni equation (Equation 2), yielding the following parameters:

$$E_g(T) = 2.459 - \frac{0.00059T^2}{33.5 + T} (3)$$

To determine the nanowire temperature with different waveguiding power, we measured the photoluminescence spectra of a CdS nanowire while increasing the power of the waveguiding CW light at 1550 nm (Figure S4c). The photoluminescence peak shows a significant red shift as the waveguiding power increases. Figure S4d plots the corresponding change in the CdS bandgap as a function of the waveguiding power (black curve). Using Equation 3, the nanowire temperature was derived from this bandgap shift as the waveguiding powers changes (red curve). For a waveguiding power of 1.1 W, the CdS nanowire temperature is about 180 °C, which is not high enough to induce observable hysteresis responses or cause damage to the nanowire.

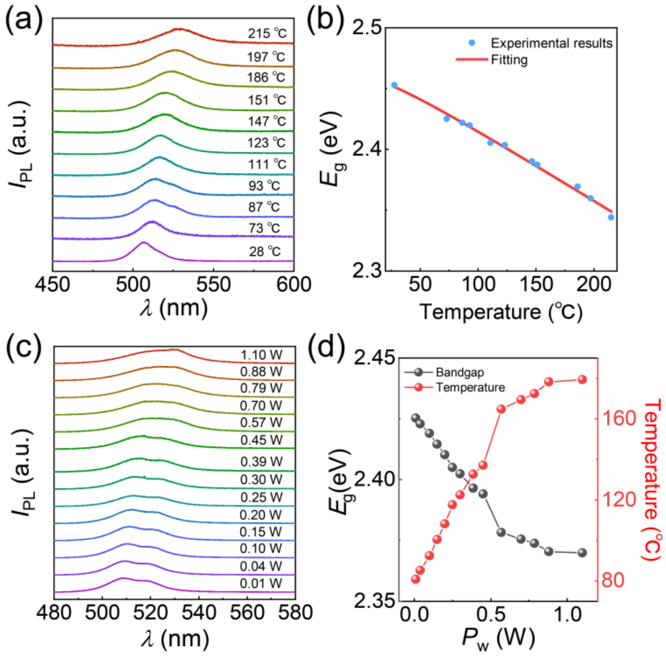

Figure S4. Nanowire temperature characterization via the Varshni effect. (a) Photoluminescence spectra of a 550-nm-diameter CdS nanowire under 355 nm pulsed excitation at different temperatures. (b) Temperature-dependent bandgap (blue dots) retrieved from (a), with a fitted line based on the Varshni equation (solid red line). (c) Waveguiding power-dependent photoluminescence spectra of a 600-nm-diameter CdS nanowire at 1550 nm. (d) Waveguiding power dependence of the corresponding bandgap (black) and derived temperature (red).

S5 | Mode analysis of the CdS nanowire

The effective refractive index n_{eff} of the first four lowest-order modes in a CdS nanowire at 1550 nm was calculated as a function of the nanowire diameter (Figure S5). The inset shows a typical field distribution of the fundamental HE_{11} mode for a 550-nm-diameter nanowire. As shown, only the HE_{11} mode is supported when the nanowire diameter is smaller than about 600 nm, satisfying the condition for single-mode operation.

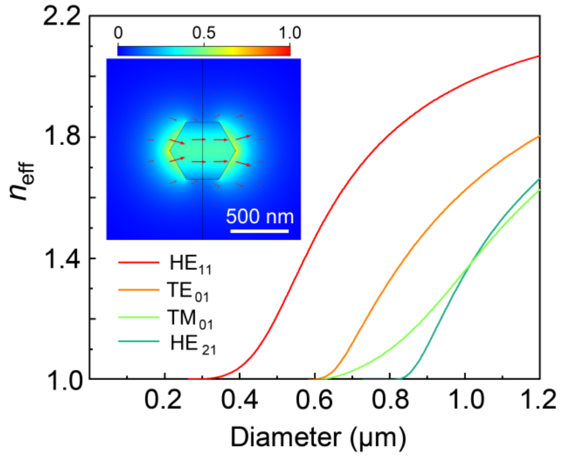

Figure S5. Mode analysis of the CdS nanowire. Diameter-dependent n_{eff} of the four lowest modes of a CdS nanowire at 1550-nm wavelength. Inset, normalized electric field distribution of the fundamental HE_{11} mode for a 550-nm-diameter nanowire.

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